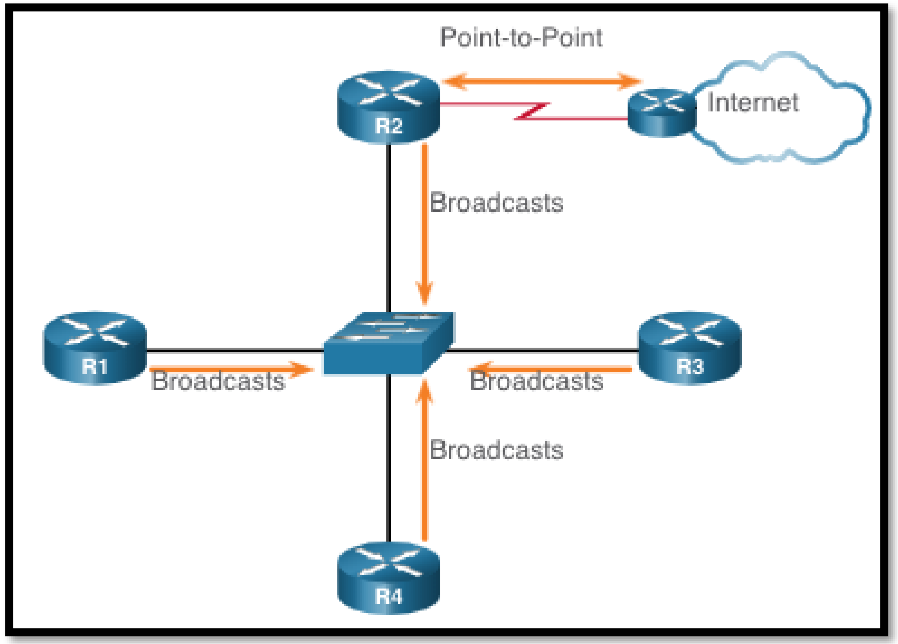
Multiaccess OSPF Networks

OPSF Network Types

- Another type of network that uses OSPF is the multiaccess OSPF network.
- Multiaccess OSPF networks are unique in that one router controls the distribution of LSAs.
- The router that is elected for this role should be determined by the network administrator through proper configuration.



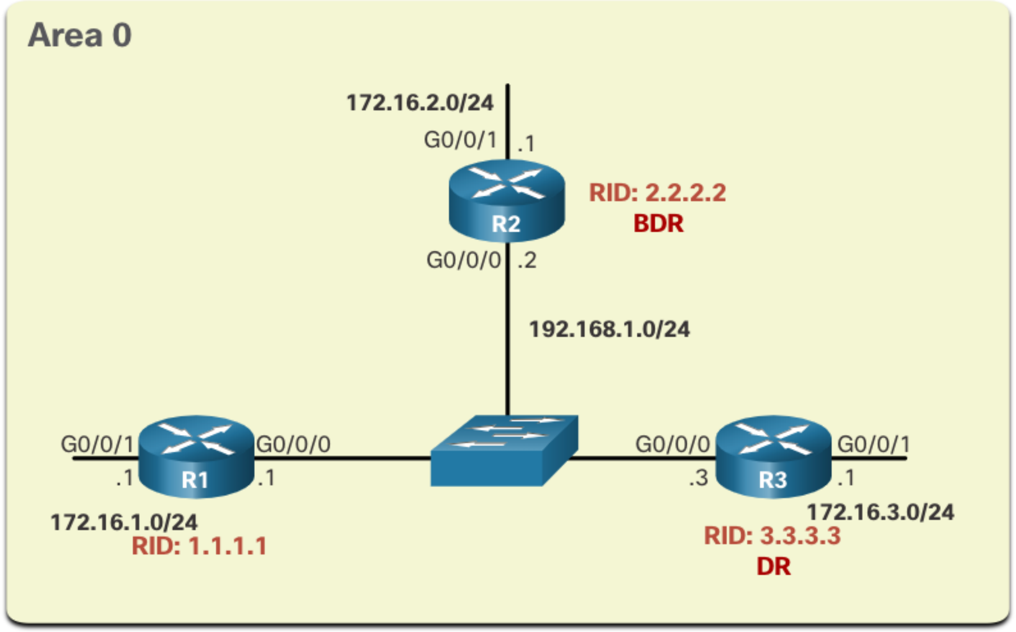
Multiaccess OSPF Networks

OPSF Designated Router

- In multiaccess networks, **OSPF** elects a DR and BDR. The DR is responsible for collecting and distributing LSAs sent and received. The DR uses the multicast IPv4 address 224.0.0.5 which is meant for all OSPF routers.
- A BDR is also elected in case the DR fails. The BDR listens passively and maintains a relationship with all the routers. If the DR stops producing Hello packets, the BDR promotes itself and assumes the role of DR.
- All other routers become a DROTHER (a router that is neither the DR nor the BDR).
- DROTHERs use the multiaccess address 224.0.0.6 (all designated routers) to send OSPF packets to the DR and BDR. Only the DR and BDR listen for 224.0.0.6.

OPSF Multiaccess Reference Topology

- In the multiaccess topology shown in the figure, there are three routers interconnected over a common Ethernet multiaccess network, 192.168.1.0/24.
- Because the routers are connected over a common multiaccess network, OSPF has automatically elected a DR and BDR.



- R3 has been elected as the DR because its router ID is 3.3.3.3, which is the highest in this network.
- R2 is the BDR because it has the second highest router ID in the network.

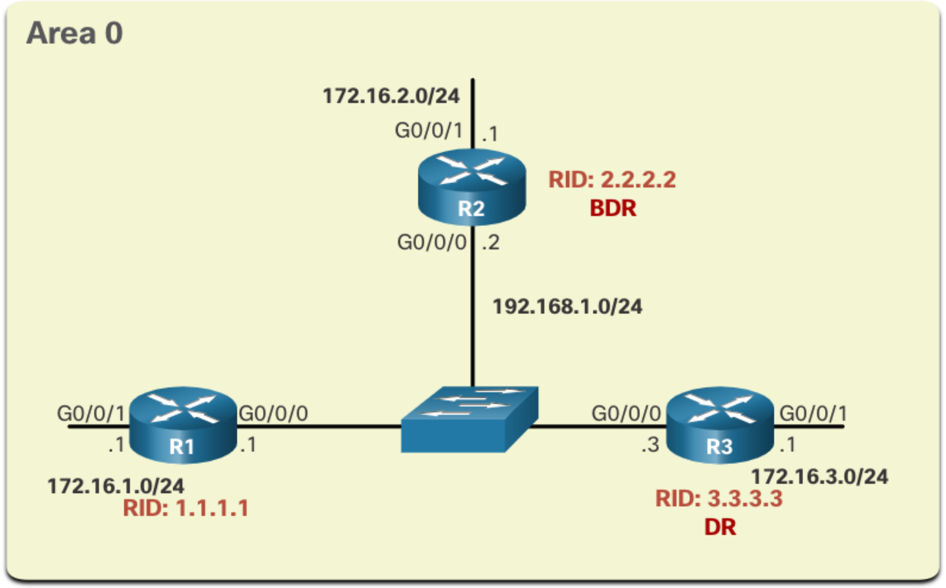
Verify OSPF Router Roles

To verify the roles of the OSPFv2 router, use the **show ip ospf interface** command.

The output generated by R1 confirms that the following:

- R1 is not the DR or BDR, but is a DROTHER with a default priority of 1. (Line 6)
- DR is R3 with router ID 3.3.3.3 at IPv4 address 192.168.1.3.
- BDR is R2 with router ID 2.2.2.2 at IPv4 address 192.168.1.2. (Lines 7 and 8)
- R1 has two adjacencies: one with the BDR and one with the DR.

```
R1# show ip ospf interface GigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet Address 192.168.1.1/24, Area 0, Attached via Interface Enable
  Process ID 10, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  (output omitted)
  Transmit Delay is 1 sec, State DROTHER, Priority 1
  Designated Router (ID) 3.3.3.3, Interface address 192.168.1.3
  Backup Designated router (ID) 2.2.2.2, Interface address 192.168.1.2
  (output omitted)
  Neighbor Count is 2, Adjacent neighbor count is 2
  Adjacent with neighbor 2.2.2.2 (Backup Designated Router)
  Adjacent with neighbor 3.3.3.3 (Designated Router)
  Suppress hello for 0 neighbor(s)
R1#
```



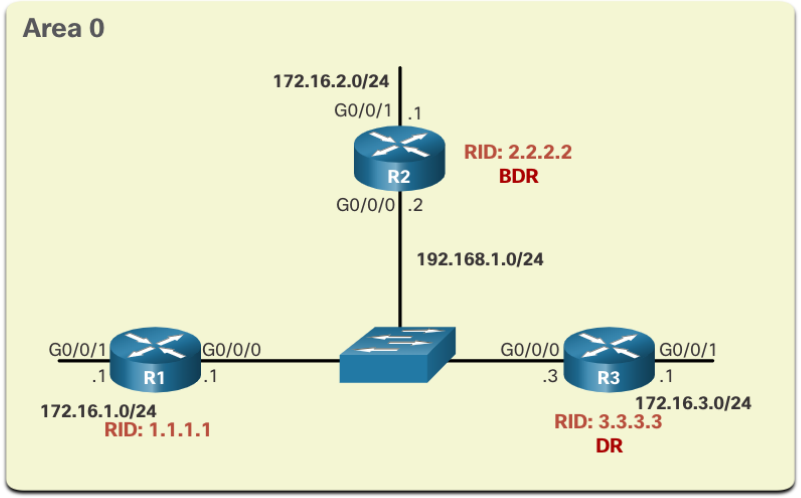
Multiaccess OSPF Networks

Verify OSPF Router Roles (Cont.)

The output generated by R2 confirms that:

- R2 is the BDR with a default priority of 1. (Line 6)
- DR is R3 with router ID 3.3.3.3 at IPv4 address 192.168.1.3.
- BDR is R2 with router ID 2.2.2.2 at IPv4 address 192.168.1.2. (Lines 7 and 8)
- R2 has two adjacencies; one with a neighbor with router ID 1.1.1.1 (R1) and the other with the DR.

```
R2# show ip ospf interface GigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet Address 192.168.1.2/24, Area 0, Attached via Interface Enable
  Process ID 10, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 1
  (output omitted)
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 3.3.3.3, Interface address 192.168.1.3
  Backup Designated Router (ID) 2.2.2.2, Interface address 192.168.1.2
  (output omitted)
  Neighbor Count is 2, Adjacent neighbor count is 2
  Adjacent with neighbor 1.1.1.1
  Adjacent with neighbor 3.3.3.3 (Designated Router)
  Suppress hello for 0 neighbor(s)
R2#
```



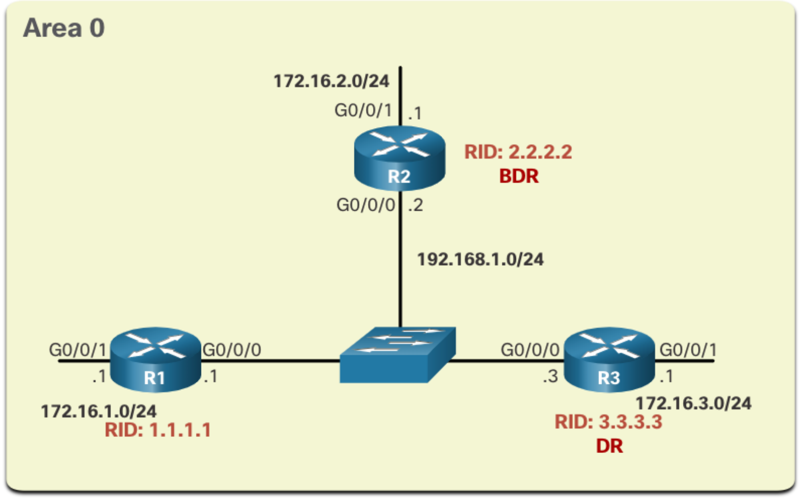
Multiaccess OSPF Networks

Verify OSPF Router Roles (Cont.)

The output generated by R3 confirms that:

- R3 is the DR with a default priority of 1. (Line 7)
- The DR is R3 with router ID 3.3.3.3 at IPv4 address 192.168.1.3.
- The BDR is R2 with router ID 2.2.2.2 at IPv4 address 192.168.1.2. (Lines 8 and 9)
- R3 has two adjacencies: one with a neighbor with router ID 1.1.1.1 (R1) and the other with the BDR.

```
R3# show ip ospf interface GigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet Address 192.168.1.3/24, Area 0, Attached via Interface Enable
  Process ID 10, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 1
  (output omitted)
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 3.3.3.3, Interface address 192.168.1.3
  Backup Designated Router (ID) 2.2.2.2, Interface address 192.168.1.2
  (output omitted)
  Neighbor Count is 2, Adjacent neighbor count is 2
  Adjacent with neighbor 1.1.1.1
  Adjacent with neighbor 2.2.2.2 (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
R3#
```



Multiaccess OSPF Networks

Verify DR/BDR Adjacencies

To verify the OSPFv2 adjacencies, use the **show ip ospf neighbor** command. The state of neighbors in multiaccess networks can be as follows:

- **FULL/DROTHER** - This is a DR or BDR router that is fully adjacent with a non-DR or BDR router. These two neighbors can exchange Hello packets, updates, queries, replies, and acknowledgments.
- **FULL/DR** - The router is fully adjacent with the indicated DR neighbor. These two neighbors can exchange Hello packets, updates, queries, replies, and acknowledgments.
- **FULL/BDR** - The router is fully adjacent with the indicated BDR neighbor. These two neighbors can exchange Hello packets, updates, queries, replies, and acknowledgments.
- **2-WAY/DROTHER** - The non-DR or BDR router has a neighbor relationship with another non-DR or BDR router. These two neighbors exchange Hello packets.

Verify DR/BDR Adjacencies (Cont.)

The output generated by R2 confirms that R2 has adjacencies with the following routers:

- R1 with router ID 1.1.1.1 is in a Full state and R1 is neither the DR nor BDR.
- R3 with router ID 3.3.3.3 is in a Full state and the role of R3 is DR.

```
R2# show ip ospf neighbor
Neighbor ID     Pri     State              Dead Time       Address            Interface
1.1.1.1           1     FULL/DROTHER       00:00:31        192.168.1.1     GigabitEthernet0/0/0
3.3.3.3           1     FULL/DR            00:00:34        192.168.1.3     GigabitEthernet0/0/0
R2#
```

Multiaccess OSPF Networks

Default DR/BDR Election Process

The OSPF DR and BDR election is based on the following criteria, in sequential order:

1. The routers in the network elect the router with the highest interface priority as the DR. The router with the second highest interface priority is becomes the BDR.
   - The priority can be configured to be any number between 0 – 255.
   - If the interface priority value is set to 0, that interface cannot be elected as DR nor BDR.
   - The default priority of multiaccess broadcast interfaces is 1.
2. If the interface priorities are equal, then the router with the highest router ID is elected the DR. The router with the second highest router ID is the BDR.
   - The election process takes place when the first router with an OSPF-enabled interface is active on the network.
   - If all of the routers on the network have not finished booting, it is possible that a router with a lower router ID becomes the DR.

DR Failure and Recovery

After the DR is elected, it remains the DR until one of the following events occurs:

- The DR fails.
- The OSPF process on the DR fails or is stopped.
- The multiaccess interface on the DR fails or is shutdown.

If the DR fails, the BDR is automatically promoted to DR.

This is the case even if another DROTHER with a higher priority or router ID is added to the network after the initial DR/BDR election.

However, after a BDR is promoted to DR, a new BDR election occurs and the DROTHER with the highest priority or router ID is elected as the new BDR.

Multiaccess OSPF Networks

The ip ospf priority Command

- If the interface priorities are equal on all routers, the router with the highest router ID is elected the DR.
- Instead of relying on the router ID, it is better to control the election by setting interface priorities. This also allows a router to be the DR in one network and a DROTHER in another.
- To set the priority of an interface, use the command **ip ospf priority** *value*, where value is 0 to 255.
  - A value of 0 does not become a DR or a BDR.
  - A value of 1 to 255 on the interface makes it more likely that the router becomes the DR or the BDR.

Multiaccess OSPF Networks

Configure OSPF Priority

- The example shows the commands being used to change the R1 G0/0/0 interface priority from 1 to 255 and then reset the OSPF process.

```
R1(config)# interface GigabitEthernet 0/0/0
R1(config-if)# ip ospf priority 255
R1(config-if)# end
R1# clear ip ospf process
Reset ALL OSPF processes? [no]: y
R1# *Jun 5 03:47:41.563: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/0/0
from FULL to DOWN, Neighbor Down: Interface down or detached
```

Lab 2.3: Determine the DR and BDR

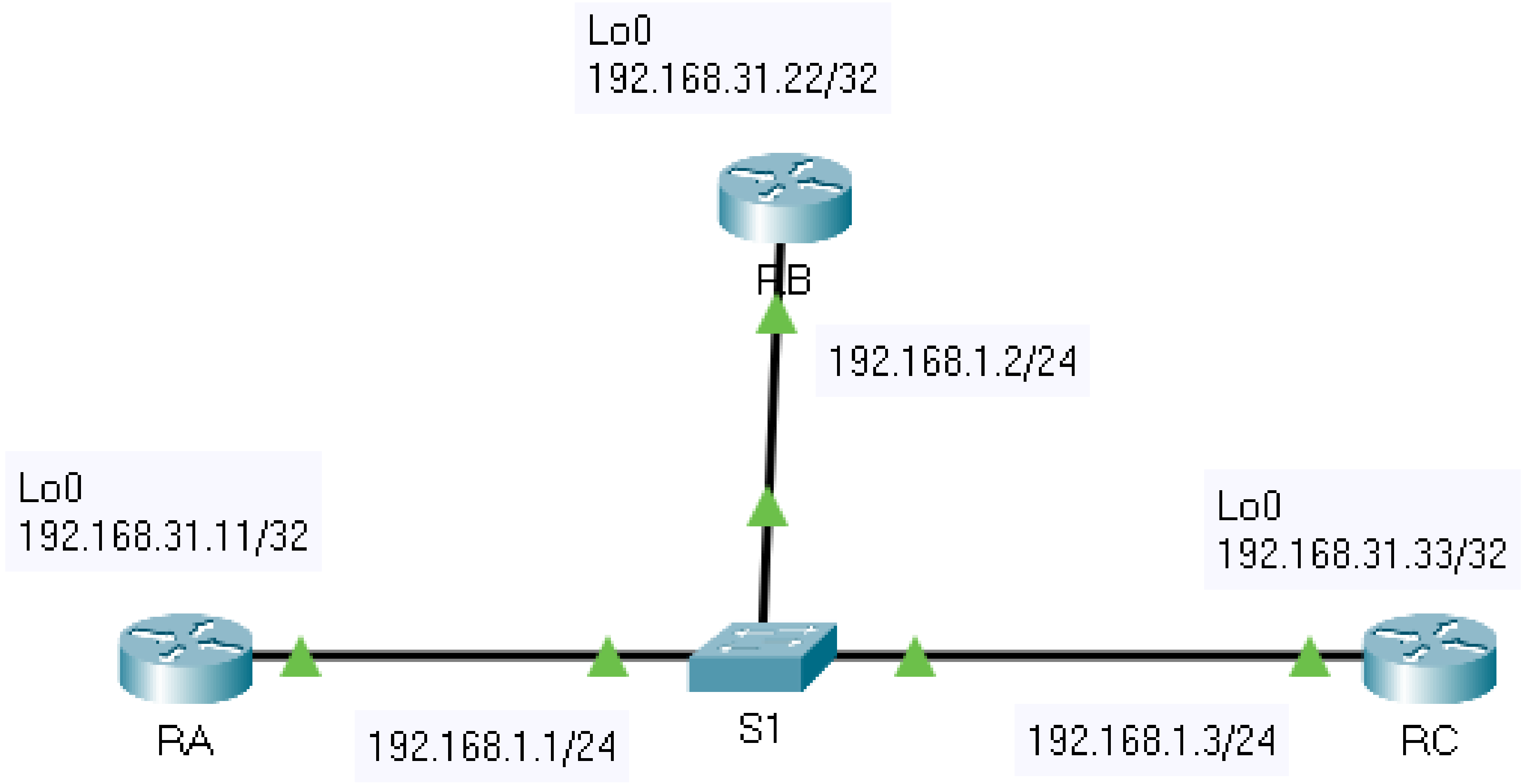
Lab 2.3: Determine the DR and BDR

Scenario

- Examine DR and BDR roles and watch the roles change when there is a change in the network.
- Modify the priority to control the roles and force a new election.
- Verify routers are filling the desired roles.

Multiaccess OSPF Networks

Lab 2.3: Determine the DR and BDR



Addressing Table

Device	Interface	IP Address	Subnet Mask
RA	G0/0	192.168.1.1	255.255.255.0
	Lo0	192.168.31.11	255.255.255.255
RB	G0/0	192.168.1.2	255.255.255.0
	Lo0	192.168.31.22	255.255.255.255
RC	G0/0	192.168.1.3	255.255.255.0
	Lo0	192.168.31.33	255.255.255.255

Multiaccess OSPF Networks

Lab 2.3: Determine the DR and BDR

Objectives

- Part 1: Examine DR and BDR Changing Roles
- Part 2: Modify OSPF Priority and Force Elections

Part 1: Examine DR and BDR Changing Roles

1. Wait until the amber link lights turn green.
2. Verify the current OSPF neighbor states.
3. Turn on IP OSPF adjacency debugging
4. Disable the Gigabit Ethernet 0/0 interface on RC.
5. Restore the Gigabit Ethernet 0/0 interface on RC.
6. Disable the Gigabit Ethernet 0/0 interface on RB.
7. Restore the Gigabit Ethernet 0/0 interface on RB.
8. Trun off Debugging.

Verify the current OSPF neighbor states

- Use the appropriate command on each router to examine the current DR and BDR.

  Example:

  R1# **show ip ospf neighbor**

- If a router shows FULL/DROTHER it means that the router is not a DR or a BDR.

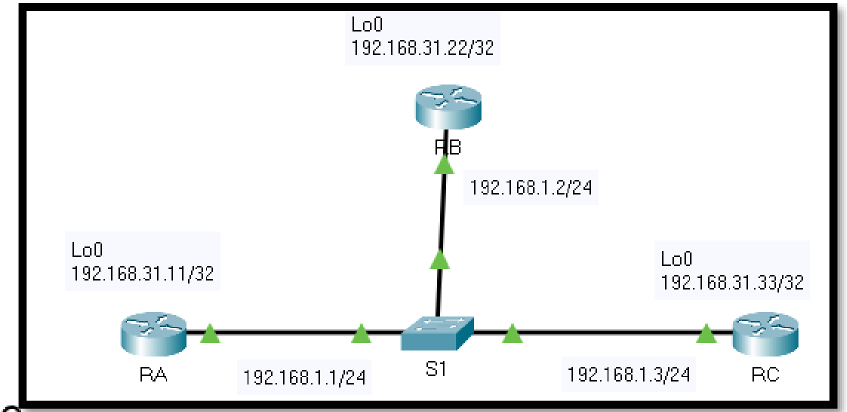
Verify the current OSPF neighbor states

RA# show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.31.33	2	FULL/DR	00:00:35	192.168.1.3	GigabitEthernet0/0
192.168.31.22	1	FULL/BDR	00:00:35	192.168.1.2	GigabitEthernet0/0

RB# show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.31.11	1	FULL/DROTHER	00:00:36	192.168.1.1	GigabitEthernet0/0
192.168.31.33	2	FULL/DR	00:00:36	192.168.1.3	GigabitEthernet0/0



RC# show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.31.11	1	FULL/DROTHER	00:00:39	192.168.1.1	GigabitEthernet0/0
192.168.31.22	1	FULL/BDR	00:00:38	192.168.1.2	GigabitEthernet0/0

Verify the current OSPF neighbor states

Which router is the DR?

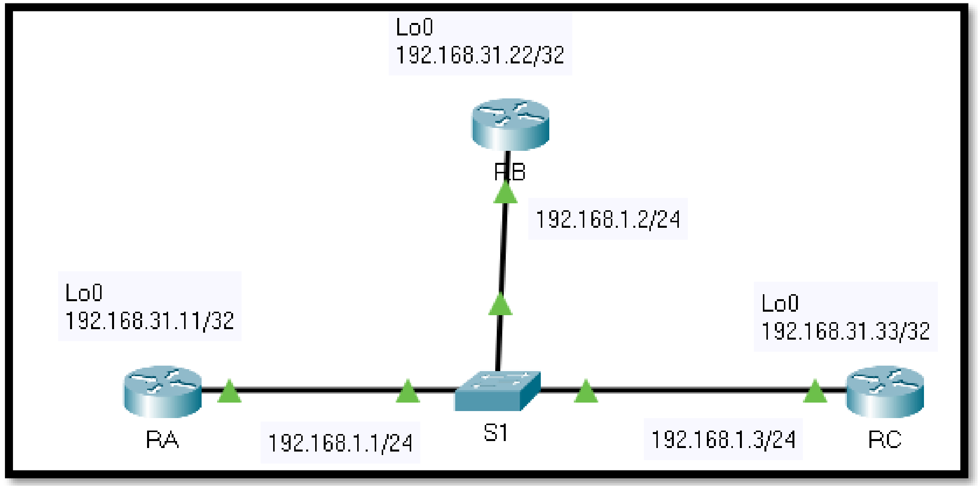
RC

Which router is the BDR?

RB

What is the OSPF state of router RA?

DROTHER



Turn on IP OSPF adjacency debugging

RA# **debug** ip ospf adj

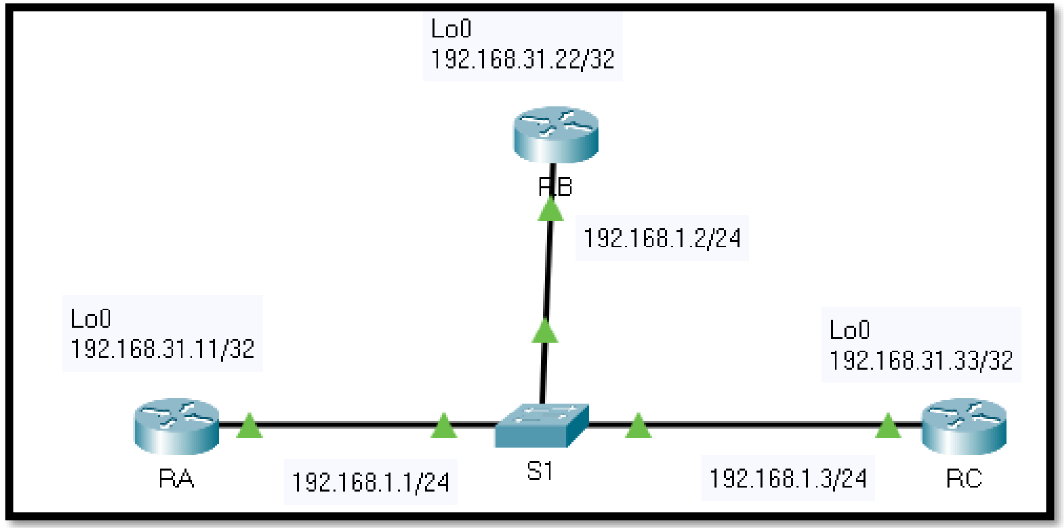
RB# **debug** ip ospf adj

Disable the Gigabit Ethernet 0/0 interface on RC.

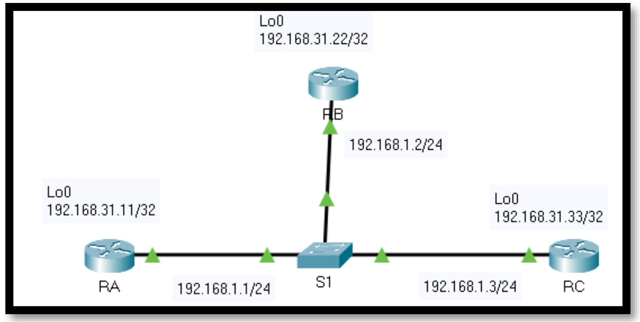
Use the **shutdown** command to disable the link between RC and the switch to cause roles to change

RC(config) #interface g0/0

RC(config-if) #shutdown



Roles change



Which router was elected DR and which router was elected BDR?

RB is now DR and RA is now BDR.

```
RB#show ip ospf interface g0/0

GigabitEthernet0/0 is up, line protocol is up
  Internet address is 192.168.1.2/24, Area 0
  Process ID 1, Router ID 192.168.31.22, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.31.22, Interface address 192.168.1.2
  Backup Designated Router (ID) 192.168.31.11, Interface address
192.168.1.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:09
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 192.168.31.11  (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
```

```
RA#show ip ospf interface g0/0

GigabitEthernet0/0 is up, line protocol is up
  Internet address is 192.168.1.1/24, Area 0
  Process ID 1, Router ID 192.168.31.11, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 192.168.31.22, Interface address 192.168.1.2
  Backup Designated Router (ID) 192.168.31.11, Interface address 192.168.1.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:02
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 192.168.31.22  (Designated Router)
  Suppress hello for 0 neighbor(s)
RA#
```

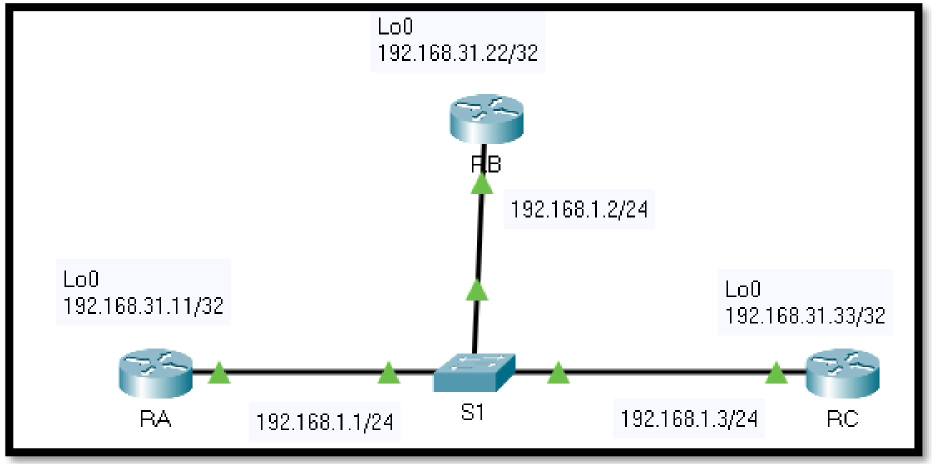
Restore the Gigabit Ethernet 0/0 interface on RC

- Re-enable the link between RC and the switch.

  RC(config) #interface g0/0

  RC(config-if) #no shutdown

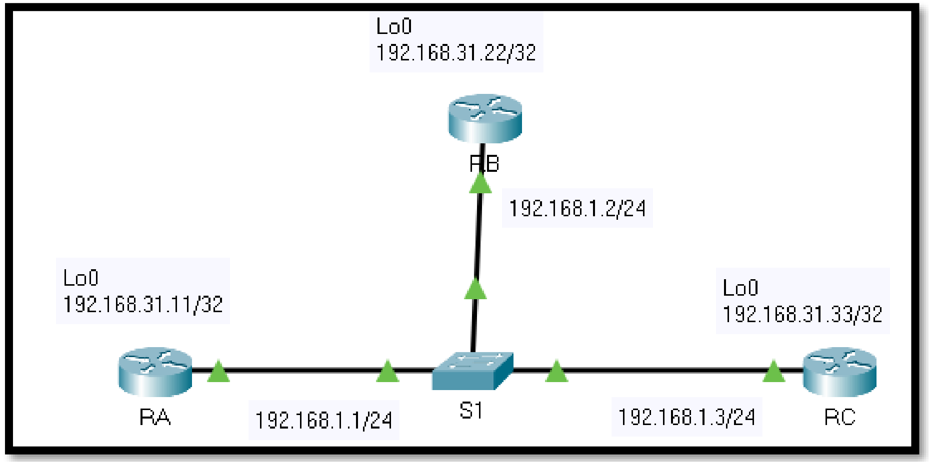
- Wait for the new DR/BDR elections to occur.



Restore the Gigabit Ethernet 0/0 interface on RC

Did DR and BDR roles change? Explain.

- No, because OSPF does not update the DR/BDR when the existing ones are active.

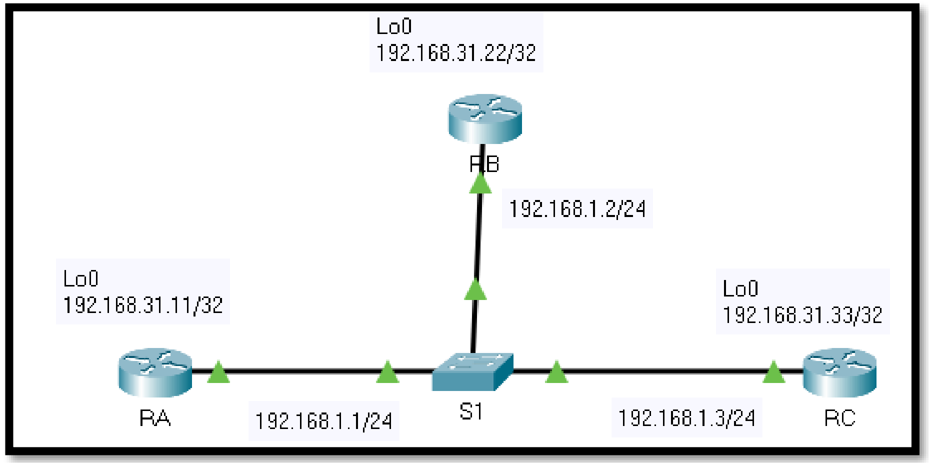


- Verify the DR and BDR assignments using the show ip ospf neighbor command on router RC.

```
RC# show ip ospf neighbor

Neighbor ID      Pri   State       Dead Time    Address        Interface

192.168.31.22      1   FULL/DR       00:00:34     192.168.1.2  GigabitEthernet0/0

192.168.31.11      1   FULL/BDR      00:00:34     192.168.1.1  GigabitEthernet0/0
```

Disable the GigabitEthernet0/0 interface on RB

Use the **shutdown** command to disable the link between RB and the switch to cause roles to change



RB(config) #interface g0/0

RB(config-if) #shutdown

Which router was elected BDR?

RA is now DR and RC is now BDR.

- (RA was the BDR, when the DR failed (RB) it became the DR.)

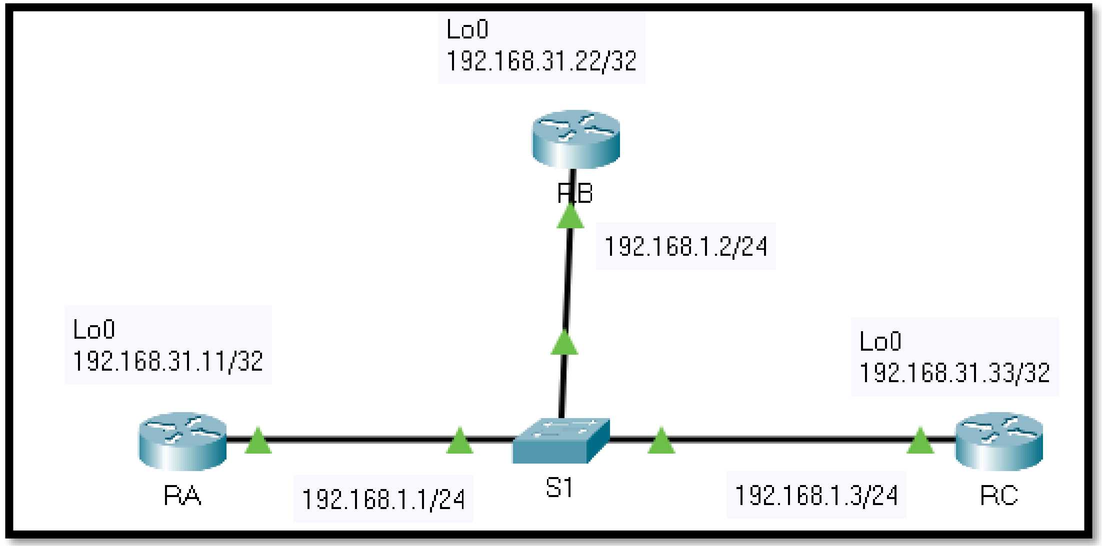
Configure OSPF priorities on each router

Set the priority on router RA, RB and RC

RA(config)# **interface g0/0**
RA(config-if)# **ip ospf priority** 200

RB(config)# **interface g0/0**
RB(config-if)# **ip ospf priority** 100

RC(config)# **interface g0/0**
RC(config-if)# **ip ospf priority** 1



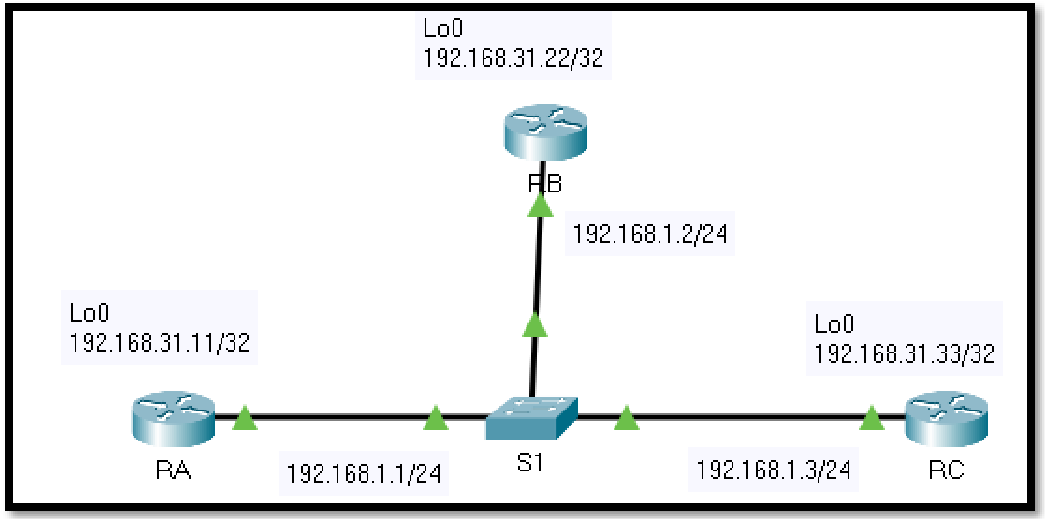
Part 2: Modify OSPF Priority and Force Elections

1. Configure **OSPF priorities** on each router.
2. Force an election by **resetting the OSPF process** on the routers.
3. **Verify DR and BDR elections** were successful.

Configure OSPF priorities on each router

a. To change the DR and BDR, use the **ip ospf priority** command to configure the GigabitEthernet 0/0 port of each router with the following OSPF interface priorities:

- RA: 200
- RB: 100
- RC: 1 (This is the default priority)



Force an election by resetting the OSPF process on the routers

Starting with router RA, issue the **clear ip ospf process** on each router to reset the OSPF process.

- Wait long enough for OSPF to converge and for the DR/BDR election to occur. This should take a few minutes.

Verify DR and BDR elections were successful

According to output from the show ip ospf neighbor command on the routers, which router is now DR and which router is now BDR?

RA is now DR and RB is now BDR.